Homework 5: Linear phase portraits and stability proofs

MATH 515 — Fall 2026

Due Friday, November 6, 2026

Assignment convention. Each numbered problem reproduces the full assigned statement from Steven H. Strogatz, *Nonlinear Dynamics and Chaos*, third edition (the specific section/problem number is given parenthetically), with some mild editing for clarity. Any additional requirements are labeled **Course-specific instructions** and are part of the assignment.

Problem 1: Attraction versus Lyapunov stability [Challenge]

Textbook problem (Strogatz, third edition, Exercise 5.1.10 and 5.1.11, "Attracting and Liapunov stable", "Stability proofs"):

For each of the following systems, determine whether the origin is attracting, Lyapunov stable, asymptotically stable, or none of the above and prove it.

(a) $\dot{x} = y$, $\dot{y} = -4x$

(b) $\dot{x} = 2y$, $\dot{y} = x$

(c) $\dot{x} = 0$, $\dot{y} = x$

(d) $\dot{x} = 0$, $\dot{y} = -y$

(e) $\dot{x} = -x$, $\dot{y} = -5y$

(f) $\dot{x} = x$, $\dot{y} = y$

Course-specific instructions. Classify all six systems and prove your classifications from the definitions. You do not need separate repetitive proofs if the arguments are similar. To disprove stability, fix one $\epsilon > 0$ and show why no $\delta > 0$ works. To disprove attraction, find initializations arbitrarily close to the origin that do not converge to it.

Problem 2: The general defective case [Challenge]

Textbook problem (Strogatz, third edition, Exercise 5.2.11):

Show that any matrix of the form

$$A = \begin{pmatrix} \lambda & b \\ 0 & \lambda \end{pmatrix}, \qquad b \neq 0,$$

has only a one-dimensional eigenspace corresponding to the eigenvalue λ. Then solve the system $\dot{\mathbf{x}} = A\mathbf{x}$ and sketch the phase portrait.

Course-specific instructions. Complete the exercise for $\lambda < 0$, $\lambda = 0$, and $\lambda > 0$. Give the explicit general solution and explain the geometry of the exceptional $\lambda = 0$ case.

Problem 3: Damped harmonic oscillator

Textbook problem (Strogatz, third edition, Exercise 5.2.13, "Damped harmonic oscillator"):

The motion of a damped harmonic oscillator is described by

$$m\ddot{x} + b\dot{x} + kx = 0,$$

where $b > 0$ is the damping constant.

(a) Rewrite the equation as a two-dimensional linear system.

(b) Classify the fixed point at the origin and sketch the phase portrait. Be sure to show all the different cases that can occur, depending on the relative sizes of the parameters.

Course-specific instructions. Assume $m > 0$ and $k > 0$.

Problem 4: Empirical classification of random linear systems

Textbook problem (Strogatz, third edition, Exercise 5.2.14(b)–(c), "A project about random systems"):

Suppose we pick a linear system at random. What's the probability that the origin will be, say, an unstable spiral? To be more specific, consider the system $\dot{\mathbf{x}} = A\mathbf{x}$, where

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Suppose we pick the entries a, b, c, d independently and at random from a uniform distribution on the interval $[-1, 1]$.

(a) Find the probabilities of all the different kinds of fixed points.

(b) To check your answers (or if you hit an analytical roadblock), try the Monte Carlo method. Generate millions of random matrices on the computer and have the machine count the relative frequency of saddles, unstable spirals, etc.

(c) Are the answers the same if you use a normal distribution instead of a uniform distribution?

Course-specific instructions. Perform the Monte Carlo portions only; an exact solution of part (a) is not required. For part (c), use independent entries from $N(0, 1)$, the standard normal law with mean 0 and variance 1. Use at least 10^5 matrices from each distribution. Give your classification rule in terms of trace, determinant, and discriminant. Report the sample size and a table of counts and estimated probabilities for saddles, stable and unstable nodes, and stable and unstable spirals. Explain why exact boundary cases such as centers have probability zero for these continuous distributions. If your numerical rule sets aside cases close to a classification boundary, state that tolerance and report their count separately. **Computational submission.** Include code that would allow another person to reproduce the calculation or plot. Clearly label every reported numerical value and plot, and explain what the output shows.